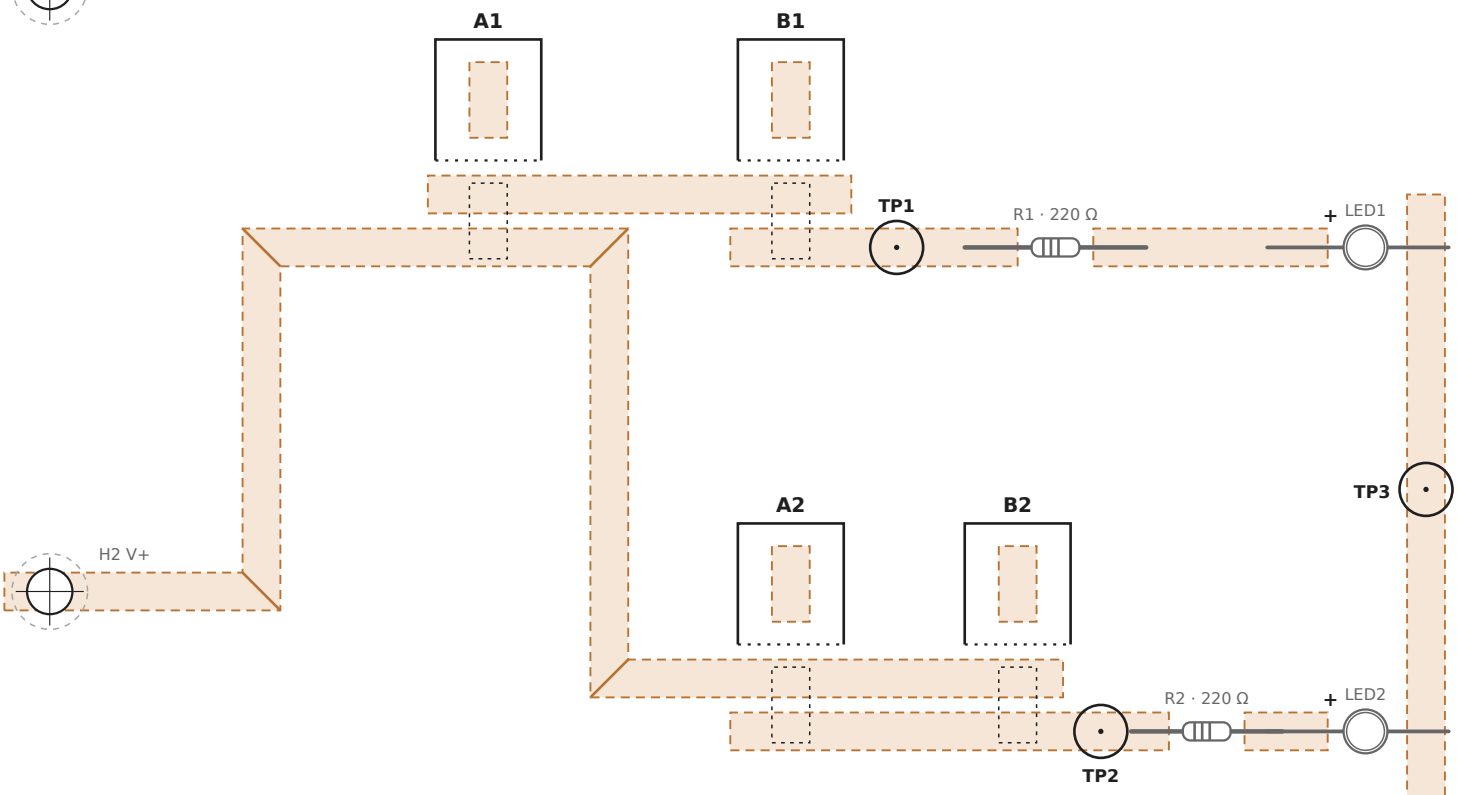
L02 Two switches, two ways

Wire the same two switches in a row, then side by side, and compare.

Draft: tagged steps may change

H1 BUS A · NC



You need

- 2 red LEDs, 5 mm
- 2 resistors, 220 Ω (red-red-brown) **[P]**
- 5 mm copper tape and scissors
- Power page P1 in the binder
- A multimeter for Test

Build it

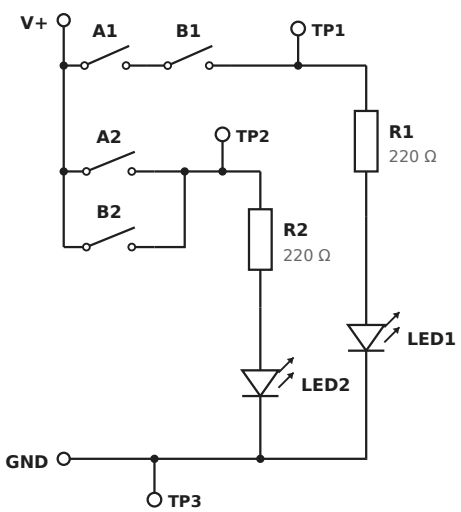
1. Lay the seven lanes, one piece of tape each. Where two lanes run close, keep the 2 mm gap clear. **[T0]**
2. Cut each flap along its solid line and crease it along the dotted hinge.
3. Stick a 10 mm piece of tape on each flap where shown. Folded down, it must cross both lanes. **[T0]**
4. Punch the holes and set eyelets in H2 and H3. **[T0]**
5. Place R1 and R2 either way round, and LED1 and LED2 as printed: long leg on the +.
6. Hold each lead flat on its lane with a small patch of copper tape. **[T0]**
7. Bind this page with P1 and switch BT1 on. Press the flaps down to test.

[T0] and **[P]** mark choices that may change: **[T0]** after the contact tests, **[P]** once the power supply is settled.

H3 GND

H4 BUS B · NC

Schematic



V+ and GND come from P1.

Predict Will the LED light? Write Y or N.

Top	LED1	Bottom	LED2
A1 only		A2 only	
B1 only		B2 only	
A1 and B1		A2 and B2	

Test Press the flaps. Black probe on TP3.

Top	LED1	TP1 V	Bottom	LED2	TP2 V
A1 only			A2 only		
B1 only			B2 only		
A1 and B1			A2 and B2		

Explain One circuit is called AND, the other OR.

Which is which? Which one is like a door with two locks?

Print at 100 %: this bar should measure 100 mm.